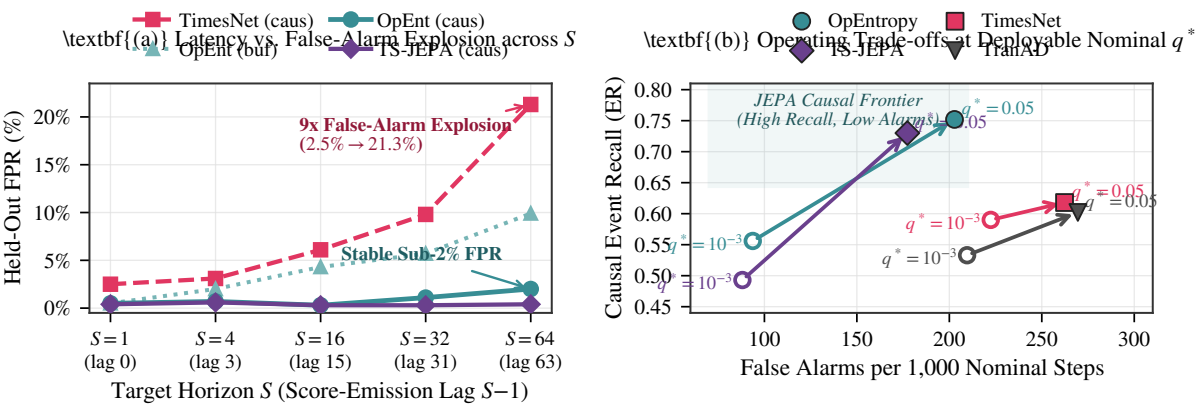
TimesNet (caus)
OpEnt (caus)
OpEnt (buf)
TS-JEPA (caus)
\textbf{(a)} Latency vs. False-Alarm Explosion across S
9x False-Alarm Explosion
(2.5% → 21.3%)
Stable Sub-2% FPR
Held-Out FPR (%)
20%
15%
10%
5%
0%
$S=1$
(lag 0)
$S=4$
(lag 3)
$S=16$
(lag 15)
$S=32$
(lag 31)
$S=64$
(lag 63)
Target Horizon S (Score-Emission Lag $S-1$)
OpEntropy
TimesNet
TS-JEPA
TranAD
\textbf{(b)} Operating Trade-offs at Deployable Nominal q^*
JEPA Causal Frontier
(High Recall, Low Alarms)
$q^* = 0.05$
$q^* = 0.05$
$q^* = 0.05$
$q^* = 0.05$
$q^* = 10^{-3}$
$q^* = 10^{-3}$
$q^* = 10^{-3}$
$q^* = 10^{-3}$
Causal Event Recall (ER)
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
100
150
200
250
300
False Alarms per 1,000 Nominal Steps